

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. AI can make mistakes, as humans it is important to be able to tell when these mistakes are made.
2. They fix errors that require an understanding of the whole codebase.
3. AI cannot sit in a meeting to figure out what a non-technical client actually wants, so a human is able to translate the messy needs of a client into technical steps.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	No	The student is primarily writing code to create a calculator. It does not involve planning, requirements, testing, etc. that stakeholders consider associated with software engineering.
B	Yes	Yes	A payroll system for 5,000 employees is a large-scale software system. It requires requirements analysis, design, testing, security, reliability, maintenance

			and consideration of many stakeholders.
C	No	Yes	The AI is generating all the code, without humans dedicated the time for it and creating the instructions for it.

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Patients	Easy access to booking information
Doctors	Access to patients records and appointment schedule
Reception Staff	Efficient patient search, appointment booking and status management
Clinic Owner	System that is cheap, efficient and easy to use while protecting patient information.

Task 2 - Identify current problems

1. Duplicate appointment bookings can occur because appointments are managed manually.
2. Patient records are difficult to locate because information is spread across spreadsheets and paper records.
3. Appointment statuses are inconsistent, making it difficult to know whether appointments are confirmed, cancelled or completed.
4. Manual processes increase the risk of errors and duplicated or outdated information.

Task 3 - Ask client questions

1. Who should be able to locate and manage patient records?
2. What information should be stored for each patient?
3. What appointment statuses do staff at the clinic need?
4. Should a patient be able to book or cancel and appointment themselves?
5. What security, backup and confidentiality of patient records does the clinic provide?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
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Appointment management	Yes – Clinic wants to improve appointment processes	Yes	Include
Facial recognition login	No – None provided on ability for clients to access information	No	Reject
AI diagnosis recommendations	No – Poor practice to perform that	No	Reject
Patient search	Yes – Streamlines ability to find patient records	Yes	Include
Online payment	No – No evidence to suggest this is an issue	No	Reject
Practitioner schedule view	Yes – The system needs to improve appointment management	Yes	Include
Insurance processing	No – No evidence to suggest this is an issue	No	Reject
Treatment-plan generation	No – Treatment plans aren't mentioned	No	Reject

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

A software engineer must verify that the software meets the client's requirements and is safe and reliable before it is used.